

Pritom Biswas

Full-Stack Developer | Integrates AI into real products | CSE undergrad, BUET

✉ pritombiswas9999@gmail.com 📞 +880 1753-861838 📍 Dhaka, Bangladesh

🌐 pritombiswas.com 🐙 github.com/Pritom2357 🔗 linkedin.com/in/pritom-biswas-11b098315



CSE undergrad at BUET who likes shipping full-stack products with AI inside. Lately: a low-cost spoken-English coach, Chrome extensions, a few hackathons and game jams, and a DL Sprint 4.0 recognition (Bengali ASR + diarization) that became a paper.

EXPERIENCE

Software Developer · Free Pixel Games Ltd.

Jan 2025 – Present

Remote / Dhaka

Node.js, PostgreSQL, React, Unity, C#, Ollama, Firebase, GTM

- Built an internal analytics tool: ask in plain English (revenue per app, session trends, crash reports by OS), get answers off game telemetry. Intent matching + query registry on Node/Postgres, scheduled reports, a locally hosted SLM via Ollama.
- Shipped *Raptor Sweep*, a Jurassic-themed Minesweeper (React + Canvas) with Firebase Remote Config A/B tests, GTM tracking and a Neon Postgres leaderboard, auto-deployed to Render.
- Day-to-day: backend features, Unity/C# gameplay work, code reviews, Git-based team workflow.

PROJECTS

Articulate AI – low-cost spoken-English coach for Bengali speakers

Jun – Jul 2026

🔗 live 🎥 demo video 🐙 articulate-ai

React 19, Node/Express 5, PostgreSQL (Neon), Azure Speech, GPT-4o-mini, FastAPI + DeepFilterNet3, Socket.io, Docker, GitHub Actions

- Speaking practice here is either a BDT 8–24k/month tutor or an app that can't hear Bangla-specific mistakes (/v/, /θ/, /æ/). This one can, at a fraction of the cost.
- Phoneme-level scoring on Azure Speech behind a DeepFilterNet3 denoiser; a transcribe-then-rescore pass makes free-speech IELTS turns scoreable; SM-2 spaced repetition driven by pronunciation score instead of self-rated recall.
- 8 chapters / 40 lessons / 7,640 words parsed from the Oxford 3000, each with IPA, Bangla meaning and TTS. Realtime via Socket.io + Postgres LISTEN/NOTIFY; CI/CD to a DigitalOcean droplet.

Google Drive Auto Downloader – Chrome extension

Dec 2025

🐙 Automatic-Google-Drive-Downloader-Extension · 3 stars

React 18 + TypeScript, Tailwind, Chrome Manifest V3, OAuth2, Node/Express, Python

- Lecture files land in five shared folders every day and Drive's own sync eats quota, so: paste a folder URL once, "Check Delta" shows only what's new, files drop into a mirrored local tree.
- Chunked downloads with pause/resume, a blacklist, runs in the background. Extension does OAuth + UI, Node does the Drive API + diffing, Python does the file I/O.

DL Sprint 4.0 – Bengali long-form ASR + speaker diarization

Feb 2026

📄 paper (arXiv:2603.03158) 🐙 repo · BUET CSE Fest 2026, hosted on Kaggle · recognized · team of 4

Whisper Medium (fine-tuned), pyannote, Demucs, VAD, Qwen post-correction, PyTorch

- Hour-long noisy Bengali recordings → who said what, when. Demucs vocal separation → VAD → chunked Whisper fine-tuned on Bengali → n-gram repetition cleanup + LLM correction; pyannote segmentation fine-tuned on the competition data for diarization.
- WER 0.38 and DER 0.27 on the private leaderboards. Wrote it up as a 5-page paper with the team.

PawPal – RAG-based pet-care platform

Dec 2025 – Jan 2026

🔗 live 🎥 demo 🐙 Paw-Protector · team of 4, built in about two weeks

React, Node.js, PostgreSQL, vector DB, Hugging Face, Tailwind

- RAG chat that answers pet-health questions against the pet's own profile, plus a custom-trained image classifier on Hugging Face for disease detection. Vet finder, vaccination reminders, adoption listings.
- I owned the system design and the AI orchestration.

Valerix – failure-tolerant microservices backend

Jan – Feb 2026

🐙 Short-Potatoes/Valerix · BUET CSE Fest 2026 Hackathon · 35 of 58 commits

Node.js, PostgreSQL ×2, Kafka, Docker Compose, Kubernetes manifests, GitHub Actions, Prometheus

- API gateway → order → inventory services, own database each, Kafka between them, deployed to a DigitalOcean box through GitHub Actions.
- Failure modes built in on purpose: "Gremlin" latency middleware, "ghost" commit-then-drop failures, and idempotent reservations so retries never double-deduct stock.

OTHER THINGS I'VE BUILT

- **RxSense** (May – Jun 2026, team, 21 merged PRs): health app with prescription reading, lab-report analysis, drug-interaction / symptom agents and RAG over MedlinePlus. Node, Postgres, LangChain, OpenAI. [repo](#)
- **AstroKrieg** (2026, IUT 12th ICT Fest GameJam, team of 3): momentum-based 2D space shooter in Unity 6 / C#: ricochet off balls, reflect enemy fire, three levels and a boss. [itch.io](#)
- **University Management System** (Mar – Apr 2026, team, about half the commits): Node/Postgres modular monolith for terms, dues, marking, with input guardrails. [repo](#)
- **SQL Data Warehouse + EDA**: Medallion (bronze → silver → gold) warehouse on SQL Server over messy CRM/ERP exports (about 60k sales rows), star schema, RFM views. [repo](#)
- **Soul** (BUET CSE Fest 2024 Game Jam, team of 3): 2D action game in Unity/C# with a “Blackhole” ability, 25+ levels and boss fights. [itch.io](#)
- **Basic-NodeJS-Template**: my backend starter (JWT + refresh, Google login, Socket.io notifications, Swagger, rate limiting), because I got tired of rewriting auth. Reused in three later projects. [repo](#)

OPEN SOURCE

- **AOSSIE-Org/PictoPy** (privacy-first desktop photo gallery, Tauri + React, 280 stars): two merged PRs, a stuck dark-mode toggle (a useEffect that kept re-writing “dark” into localStorage) and hover titles + aria-labels across the UI; prototyped webcam face-search with a device picker. [my PRs](#)

PUBLICATION

An Investigation Into Various Approaches For Bengali Long-Form Speech Transcription and Bengali Speaker Diarization. arXiv:2603.03158, Mar 2026. [arXiv](#) [ResearchGate](#)

EDUCATION

B.Sc. in Computer Science & Engineering · BUET

Aug 2024 – Present

Bangladesh University of Engineering and Technology, Dhaka. Coursework: Data Structures & Algorithms, OOP, Database Systems, Discrete Mathematics, Numerical Methods, Software Design Patterns, Computer Networks

SKILLS

Languages	JavaScript / TypeScript, Python, SQL (T-SQL, PL/SQL), C++, Java, C#
Backend	Node.js, Express, PostgreSQL (Neon), REST, JWT, Socket.io, Swagger, FastAPI, Kafka (basics)
Frontend	React, Vite, Tailwind CSS, Chrome Extensions (MV3), Tauri
AI / ML	RAG with LangChain + vector DBs (HyDE, RRF, query decomposition, semantic routing), OpenAI API, Azure Speech, Whisper fine-tuning, pyannote, Hugging Face, PyTorch, Ollama
DevOps	Docker / Compose, Kubernetes (Kind), GitHub Actions, Nginx, PM2, Prometheus, Render, DigitalOcean
Also	Unity 2D, Git, Linux, Postman

COMPETITIONS & ACTIVITIES

- **DL Sprint 4.0** (BUET CSE Fest 2026, Kaggle): recognized for Bengali long-form speech recognition + speaker diarization.
- **Hackathons & jams**: BUET CSE Fest 2026 Hackathon (Valerix), IUT 12th ICT Fest 2026 GameJam (AstroKrieg), BUET CSE Fest 2024 Game Jam (Soul).
- **Codeforces**: Pupil, rating 1308, 20 rated contests (handle: pritombiswas9999).